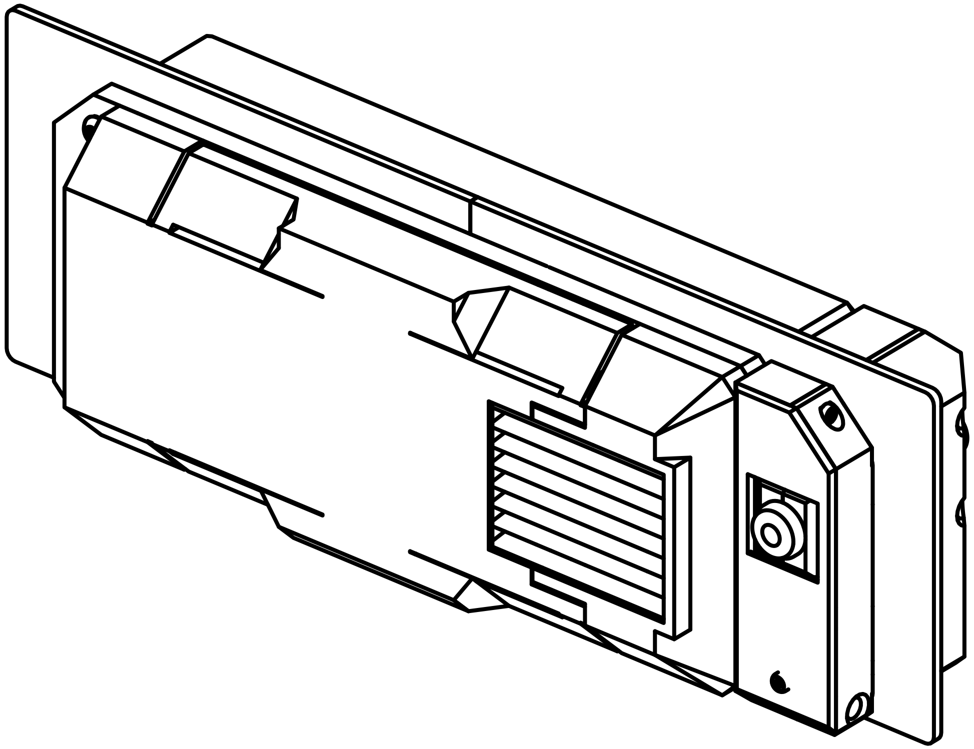
OITSWILLIAMV0

Exhaust XS

For Voron 0

Assembly manual

REV 2026-06-15



OITSWILLIAM PANG

Hardware

Contents

Before getting started

You are about to read the manual of Exhaust XS.

Warning

Danger: unless if it only runs on DC and only has a DC input, otherwise risk of electric shock

- Even though you can build with this mod from start, if you want to wire, or re-wire, you need to make sure and check that the mains is unplugged.
- Failure doing so which is unplugging the mains power may cause electrocution or even death.
- When in doubt, please don't touch the wires connected to the power supply or SSR.

Other notes

- **Exhaust XS does not support hose adapters from ELG2 or ELG1.**
- There is a reserved component that is [redacted], but we will reveal it within foreseeable future, so stay tuned by following Oitswilliam Pang in Facebook, Instagram and YouTube.

About Exhaust XS

Exhaust XS (XS stands for eXtra small, abbreviated EXS) is Oitswilliam-designed exhaust unit made for Voron 0. It is low-profile, has dual-layer HEPA filters to filtrate microplastics and activated carbon pellets to capture VOCs. EXS uses 5010 blower fan to make exhausting sound different whilst performant. It partially uses Voron Design's Stealth design scheme. EXS uses compliant design just to reduce raw materials and ease assembly, such as removable pellet panel and tool-less cartridge removal. It will pave for the next generation of EL for standard Voron printers.

Dark lore: The name EXS has been around inside Oitswilliam's brain since around 2024.

Compatibility
Exhaust XS is compatible with Voron 0, with pre-made 3D-printable panels made specifically for V0.1 (printed top-hat) and V0.2 / V0.3 (extrusion top-hat). Other printers that are smaller than 40cm in either dimensions and did not have a pre-done exhaust cutout are also applicable. Note that top-hat from Voron 0.0 is incompatible.

Contribution

This project can be contributed through GitHub. You can visit, contribute and report new issues here.
Technical support
You can contact Oitswilliam here. You can email or direct message Oitswilliam in Facebook. Or if the issue is not related to OITSWILLIAMV0 mod, but rather most of Voron parts, you may visit Voron forum or its Discord server.

Guide and legend

- **Material** - raw material
- **Material** - highlighted material
- **Model** model.stl / model.3mf - printed material, as this part is simple, the printed parts may be color-coded.

Part printing

The parts of this model can be printed in the base color of the printer. **Bold text** indicates the recommended value and *italic text* indicates on what setting / value it needs to set.

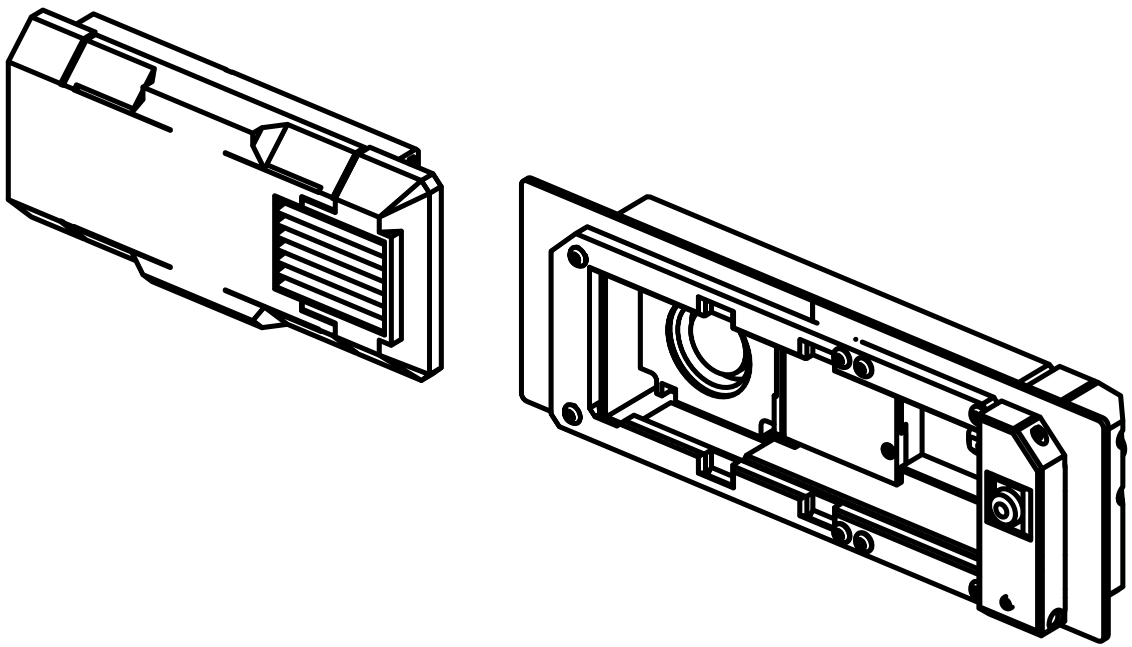
FDM / FFF or SLS	Print material: ABS, ASA or PA12, **prohibits PLA.**
Layer height of **0.2mm,** initial layer height can be **0.2mm or higher.**	*Nozzle and extrusion width* must be **forced 0.4mm; if using 0.4mm nozzle**
1.6mm *wall width*, or **4** *wall counts.*	*Infill* must be **at least 40%, except on the infill modifiers**.
At least 4 *solid bottom and top layers*, **except on the infill modifiers**.	The best *print speed* is up to **120mm/s**. The speed might be reduced by the hot end flow rate.
Supports are **recommended for main skirt parts.** Use **tree or organic supports** if available.	*Other settings might be depended on your printer.* *Follow it's warnings for proper handling.*

Additional maker tools

To get started building the project, you may use the following tools:

- Wire crimping tool - to make custom-length connectors,
- Soldering iron - to connect the switches with wires,
- Heat shrink tubings, in 2mm diameter, three colors minimum.

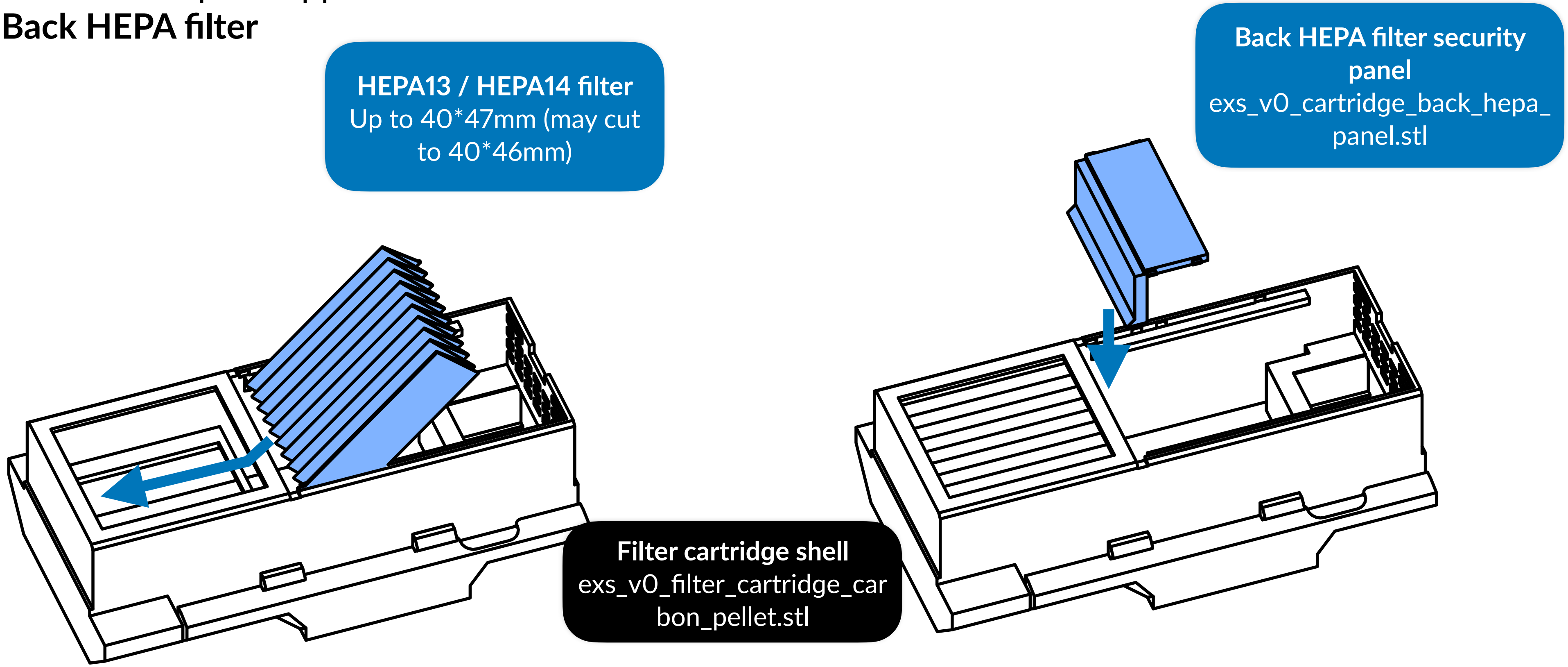
Assembly



Filter cartridge

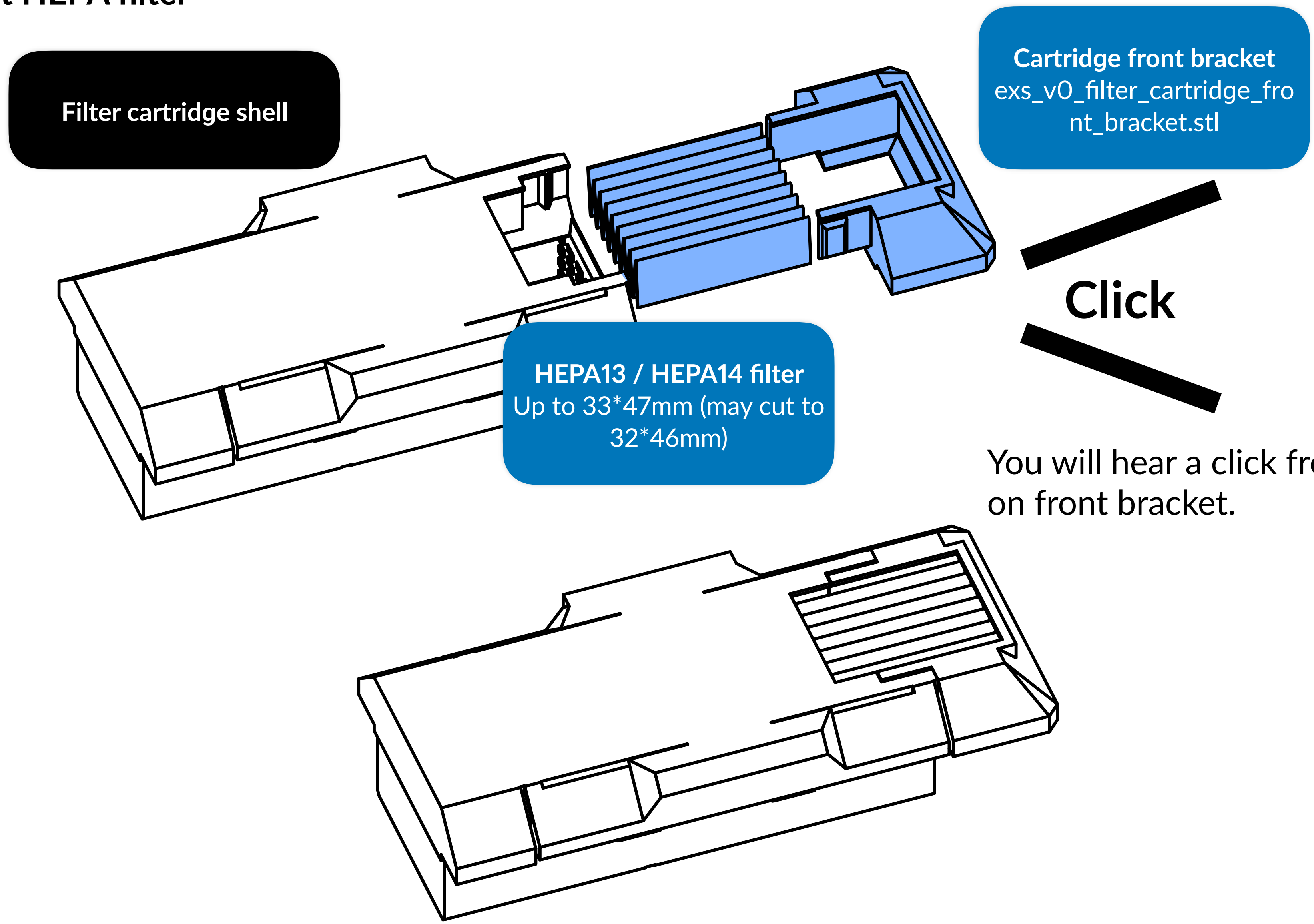
1mm foam tape wrapped around the HEPA filters' sides is recommended.

Back HEPA filter



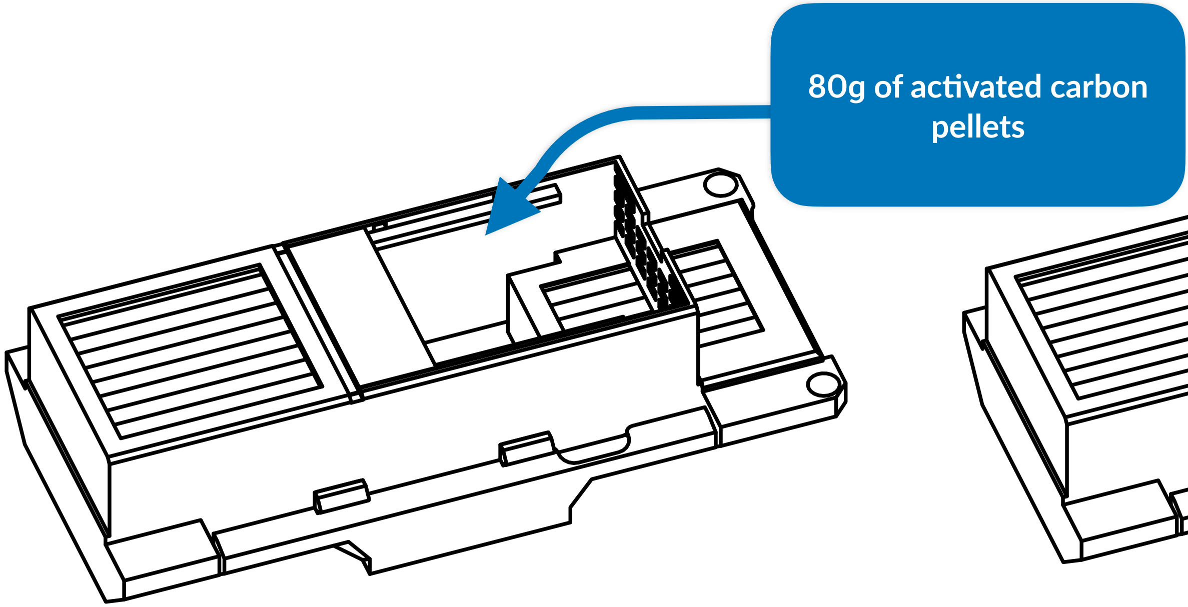
You may squeeze the filter.

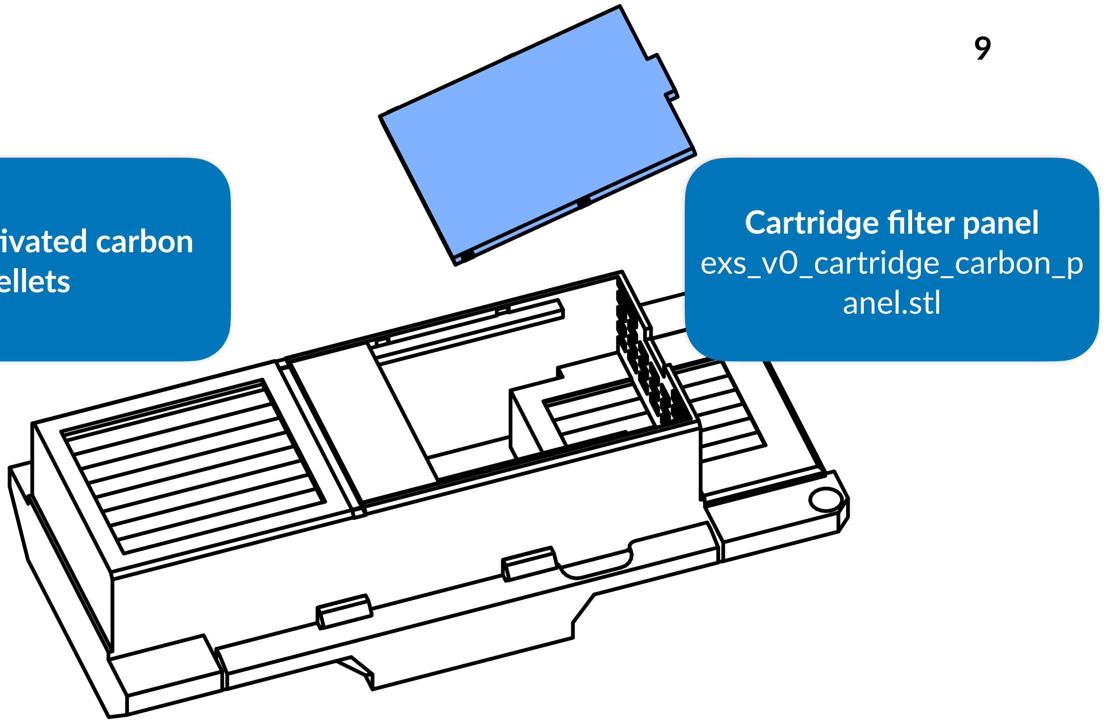
Front HEPA filter

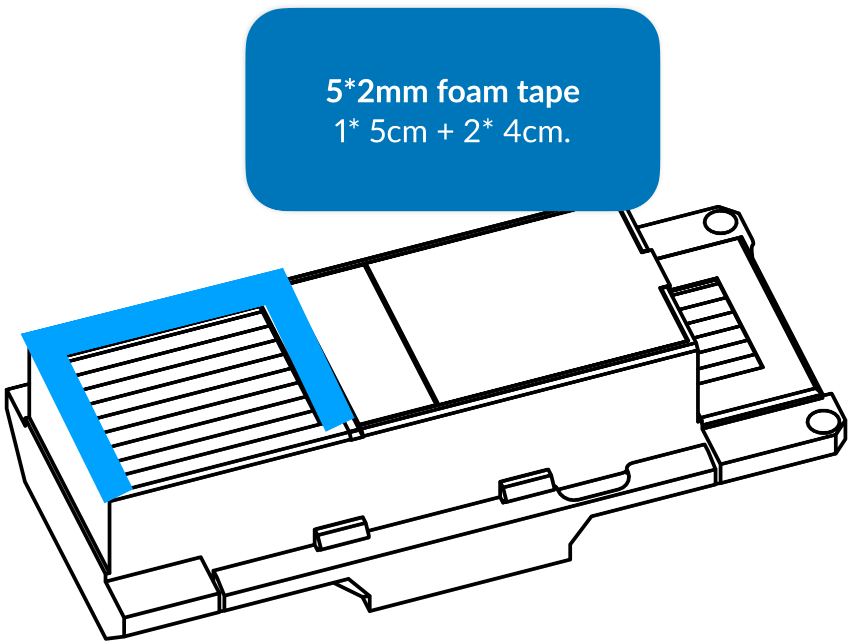


You will hear a click from a snap-on front bracket.

Activated carbon pellets





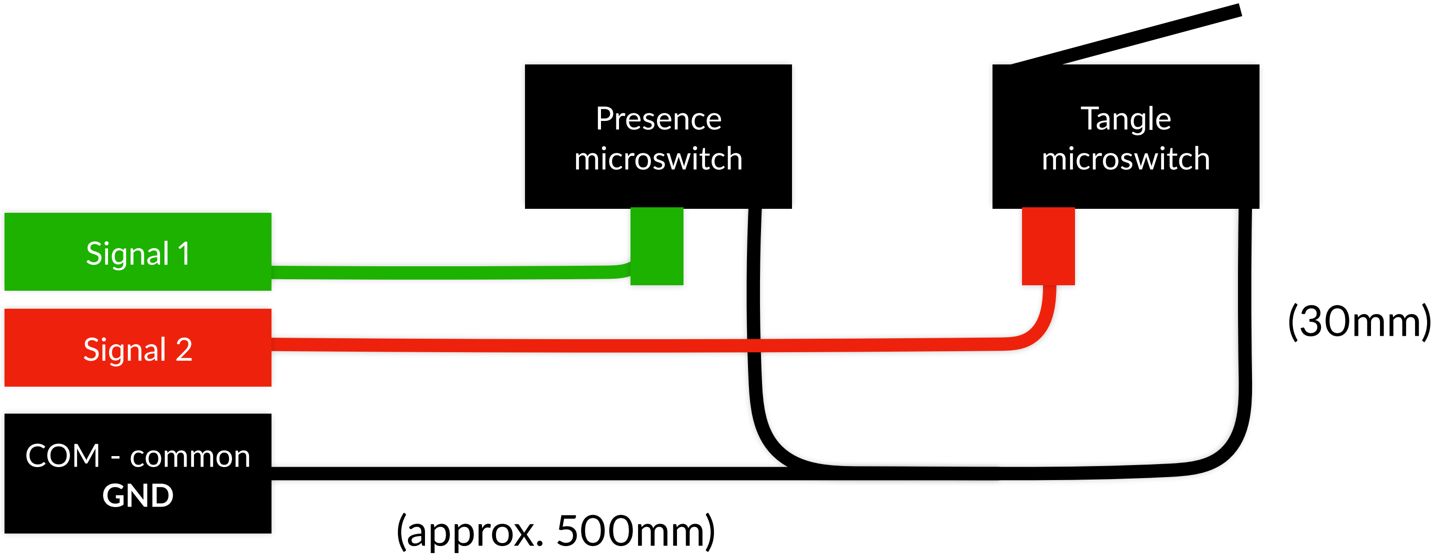


The cartridge is built.

Hybrid filament sensor (presence and tangle sensor)

This is recommended if your Voron 0 did not have any filament run-out and / or tangle sensor. EXS can be installed with or without hybrid filament sensor. ~~This can be ignored if you are installing EXS with Mellow MDM.~~

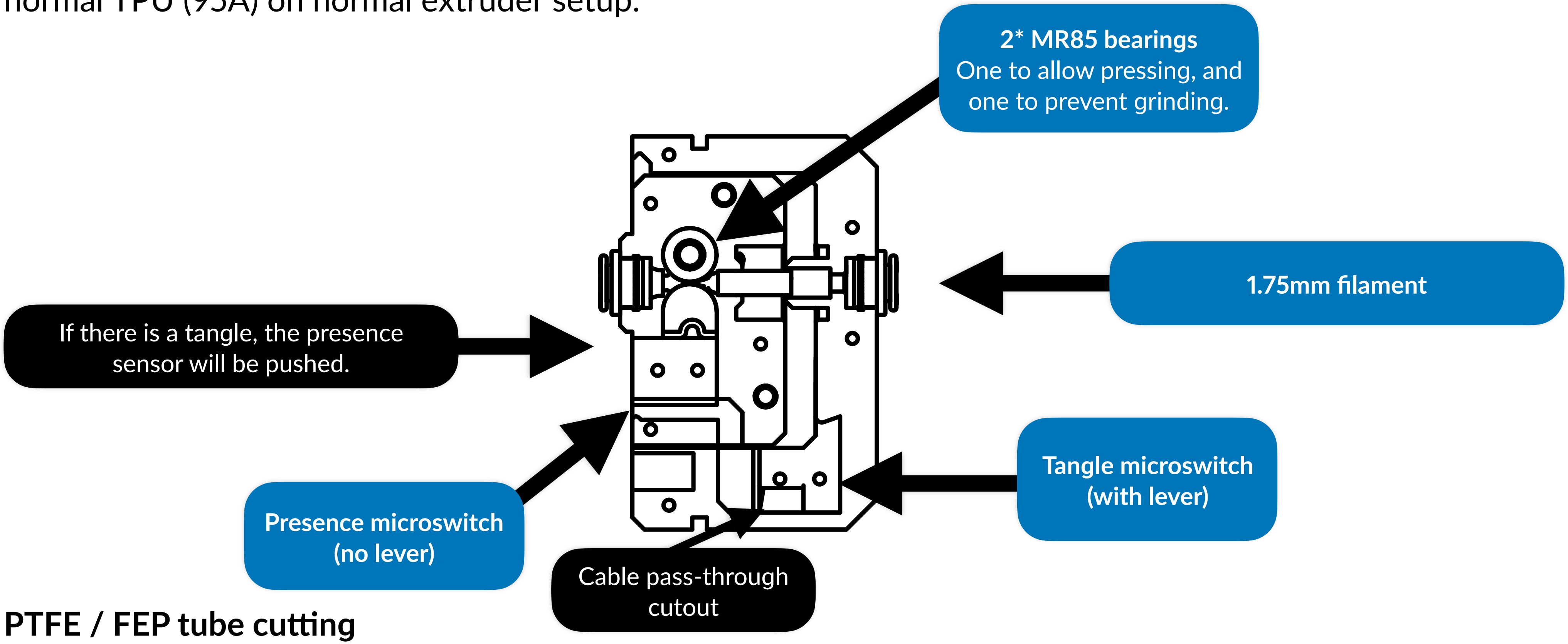
Wiring



Note: it's best time to have two pairs & three colors of heat-shrink tubes for color indication.
Connect the first long signal wire to the normally open (NO) terminal of the presence switch, and then connect the second long signal to the normally closed (NC) terminal of the tangle switch, after that, use the 30mm wire to connect the common (COM) of the switches and use the last long signal wire to the presence switch.

Anatomy of the hybrid filament sensor

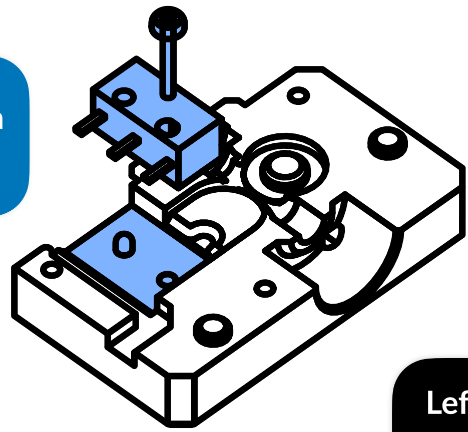
The hybrid filament sensor is made with two switches that detect respective filament-error failures, run-out and tangle. Although it does not detect jams, where the filament stops extruding due to issues caused from the HotEnd, such as heat creep or residue, it is enough to detect whether the filament tangle and the presence of the filament. Be noted that it will trigger filament tangle if the pre-presence tube setup has too much friction to deal with, and the tangle sensor can't detect materials softer than normal TPU (95A) on normal extruder setup.

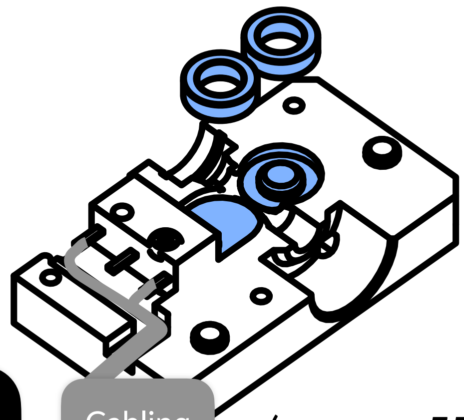


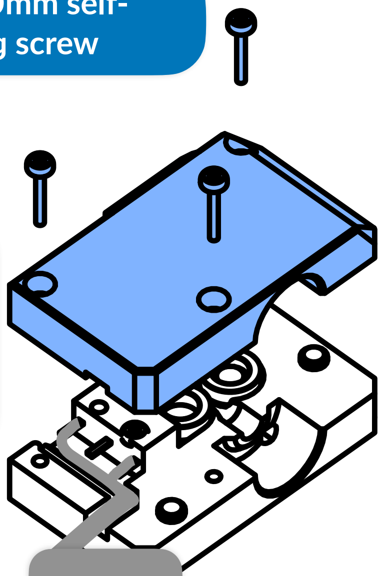
PTFE / FEP tube cutting

Cut the tube in 2 to 3mm ID to around 12mm and then trim or drill the beginning of the tube just to have the inner chamfered. The trimming / drilling guide from Prusa can be found here.

Hybrid filament sensor assembly



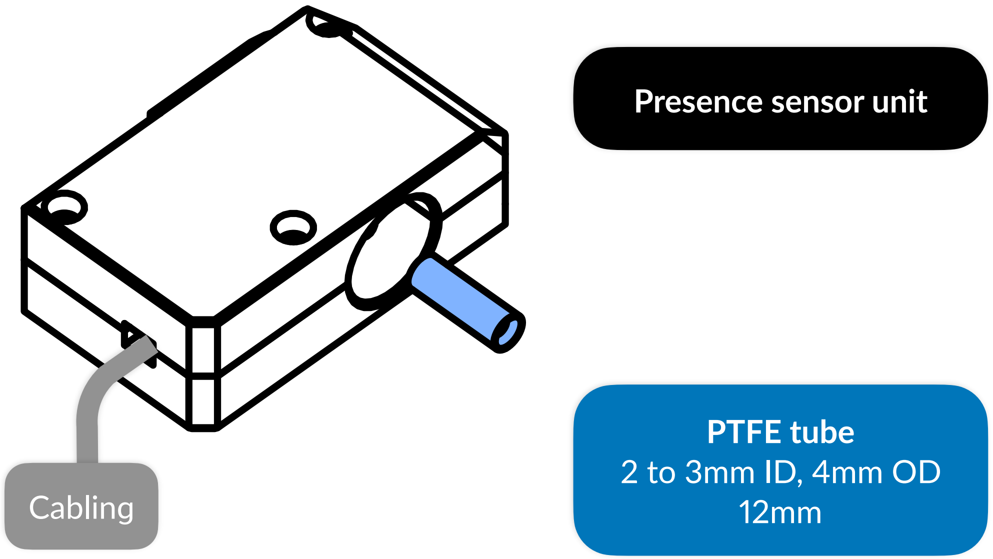




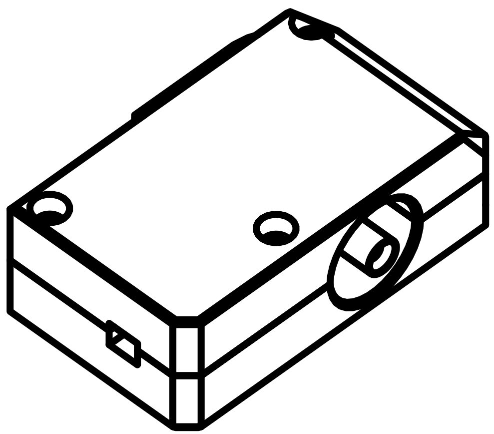
Watch out the cables wired.

ECAS04 coupler can be added now before right side of the panel or later.

Don't fully tighten the screws just yet, you will add the 12mm PTFE tube.

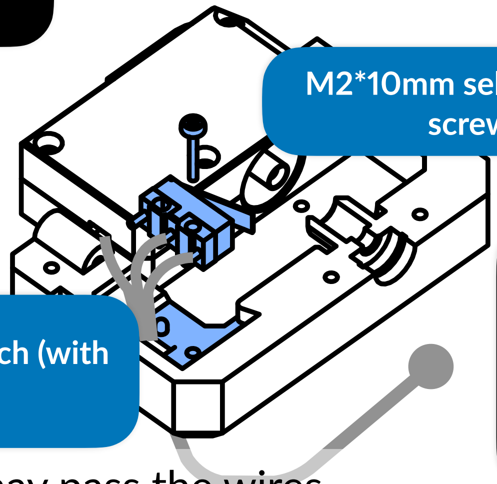


Attach the PTFE tube in to the premade hole. The chamfered side of the tube must be facing outside.

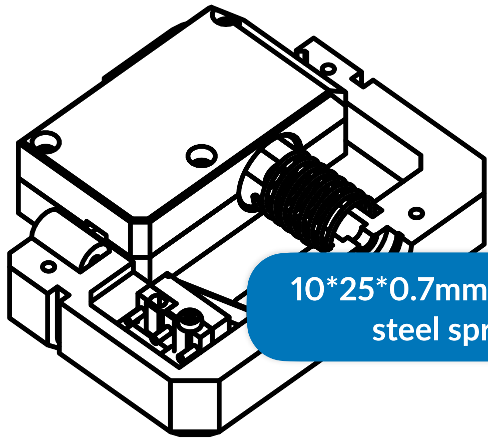


The assembly of the presence sensor is complete.

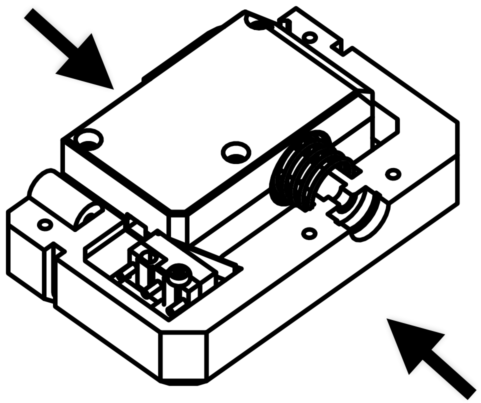
Tangle sensor



Note: you may pass the wires through the cutout near the tangle switch mount.

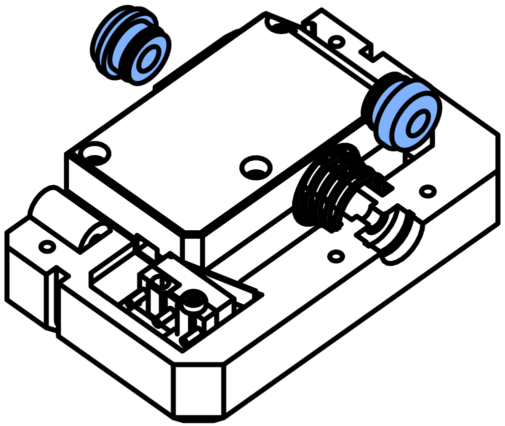


Note: watch out for the cables. You may lift the presence sensor up.



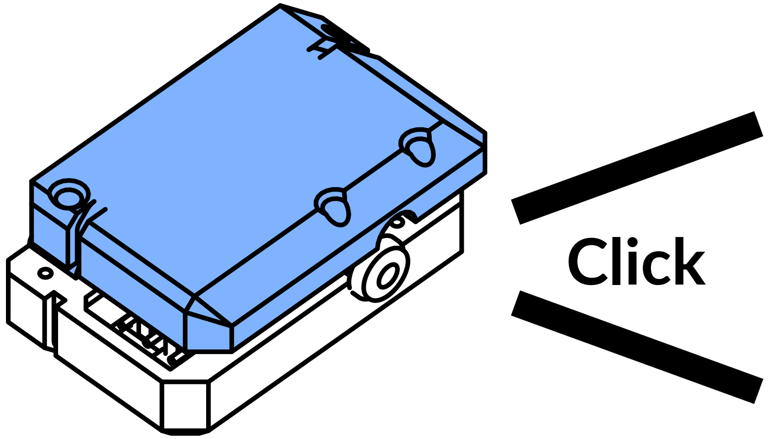
Test the tangle microswitch by pressing the presence sensor and the shell, you may put the hand at the righthand side of the presence sensor as to prevent component flyaway.
You should hear a click, if not, the switch might be incompatible or the part printed has settings badly tuned. If the problem persists, please flag it as error on GitHub.com/Bunny350/OITSWILLIAMV0/issues.

2* ECAS04 coupler

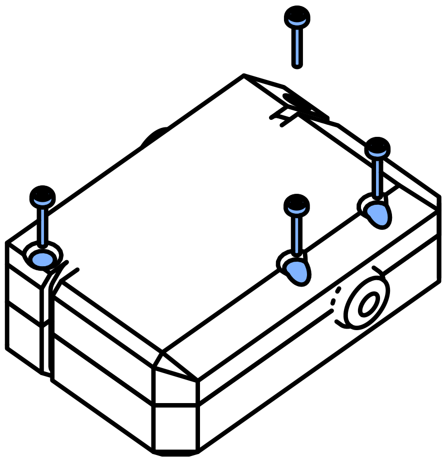


Left side of the sensor unit shell

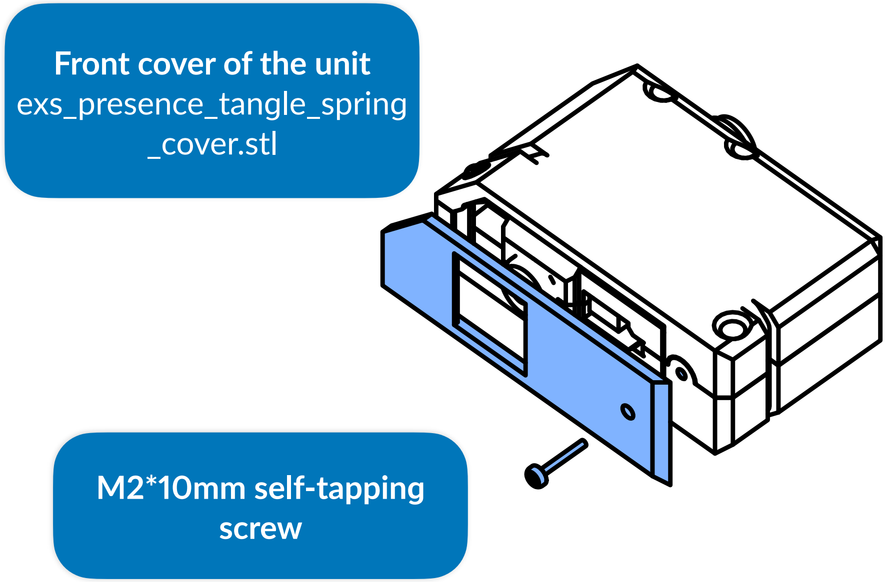
Right side of the sensor unit shell
exs_presence_tangle_shell_r.stl

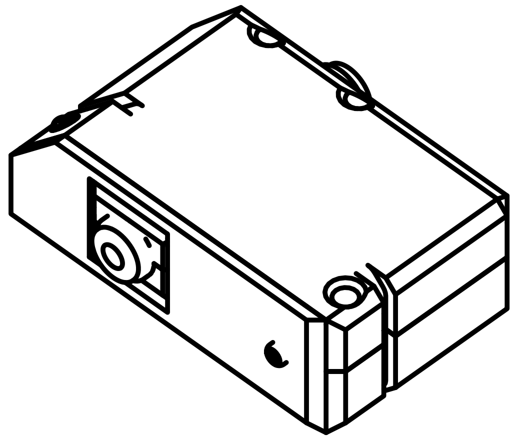


You may hear a spring click.



4* M2*10mm self-tapping screws

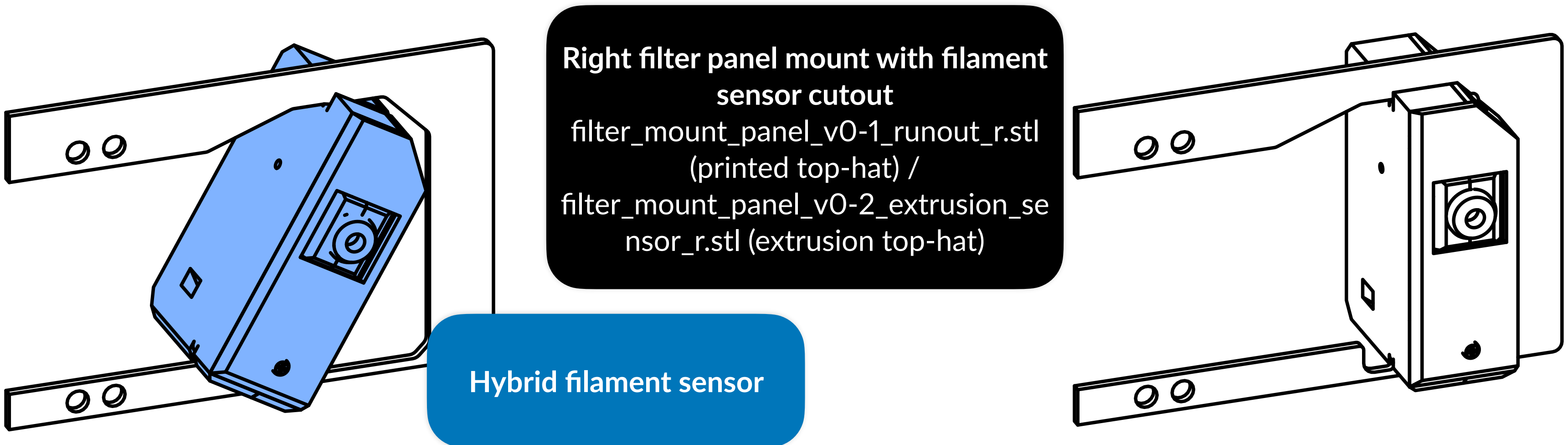




The unit is assembled.
It can now then be used
as standalone or be
attached to the panel.

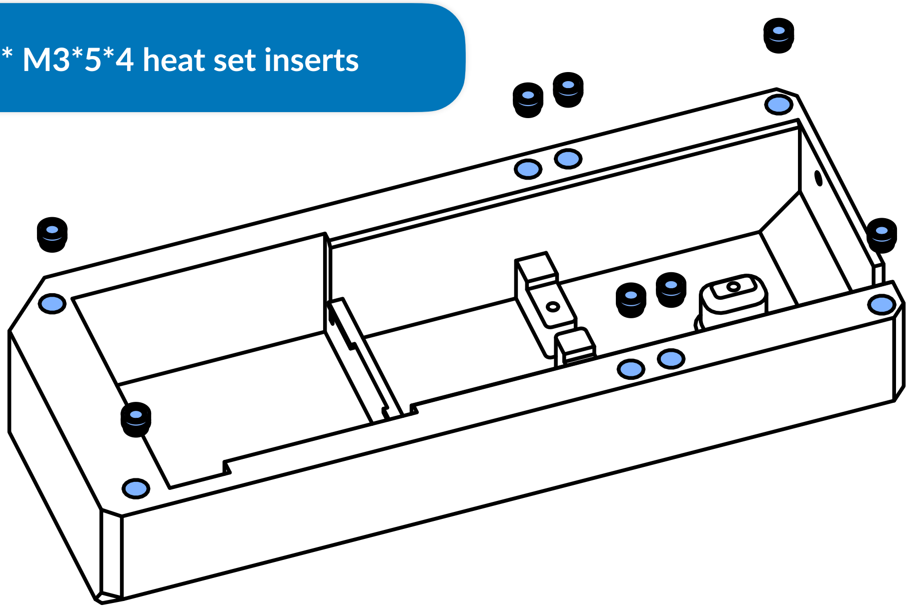
Mounting to panel

Showing panel mount for extrusion top-hat. This instruction is compatible with variants for printed top-hat. The panel may bend. Because of this, it is recommended that the panel be made in ABS / ASA. Mount the sensor tilted to prevent excessive bending.



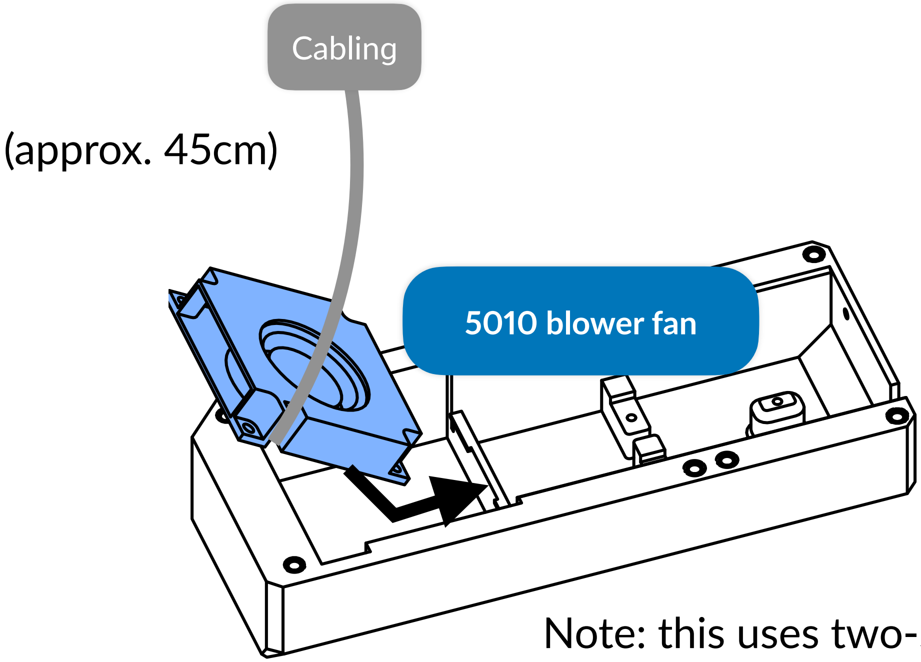
The outer side
Heat set preparation

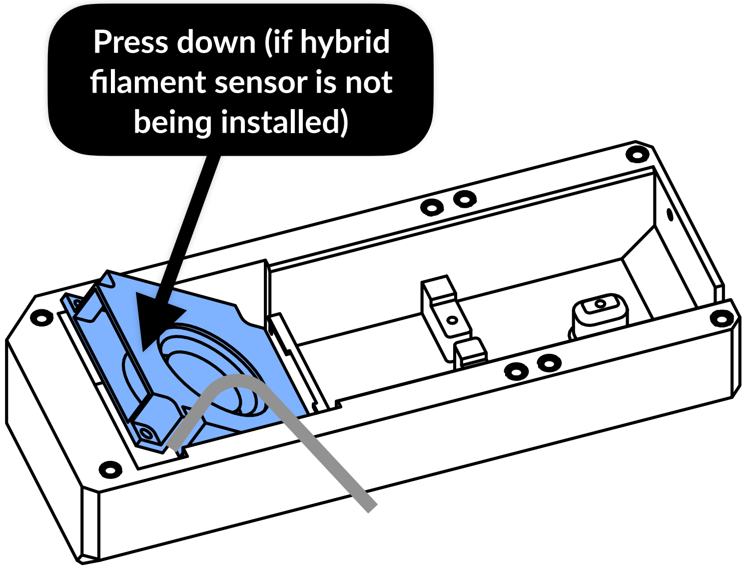
8* M3*5*4 heat set inserts



Bottom of the main unit
exs_v0_filter_main_unit_bottom.stl (large-size bed version) /
exs_v0_filter_main_unit_bottom_split_l.stl +
exs_v0_filter_main_unit_bottom_split_r.stl

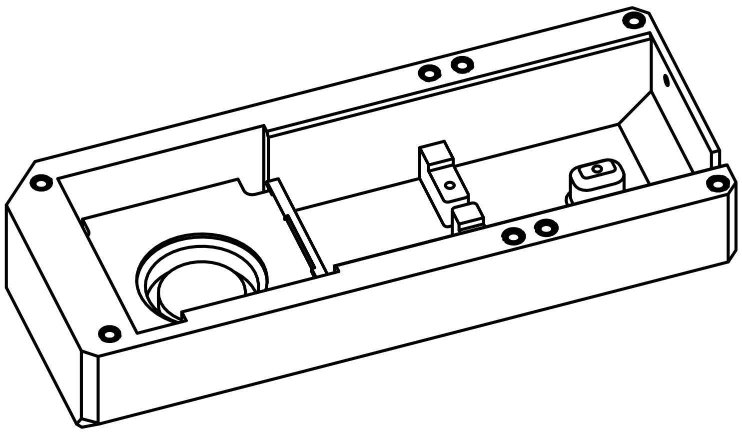
Electrical installation



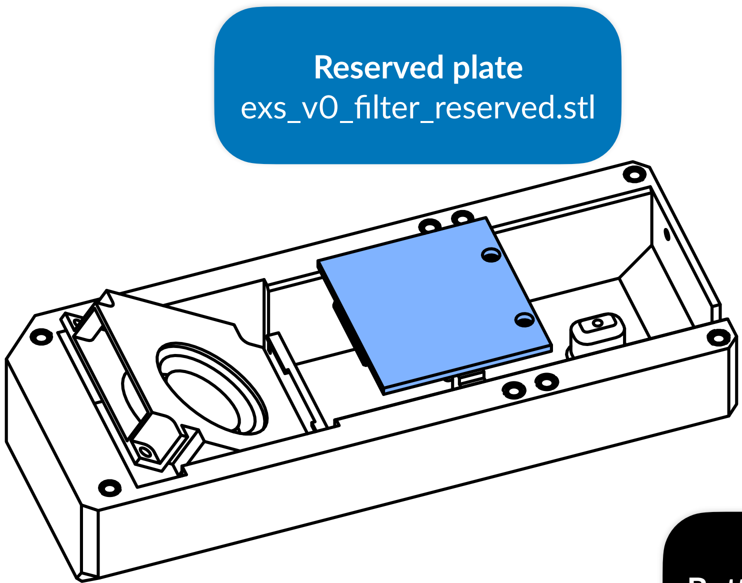


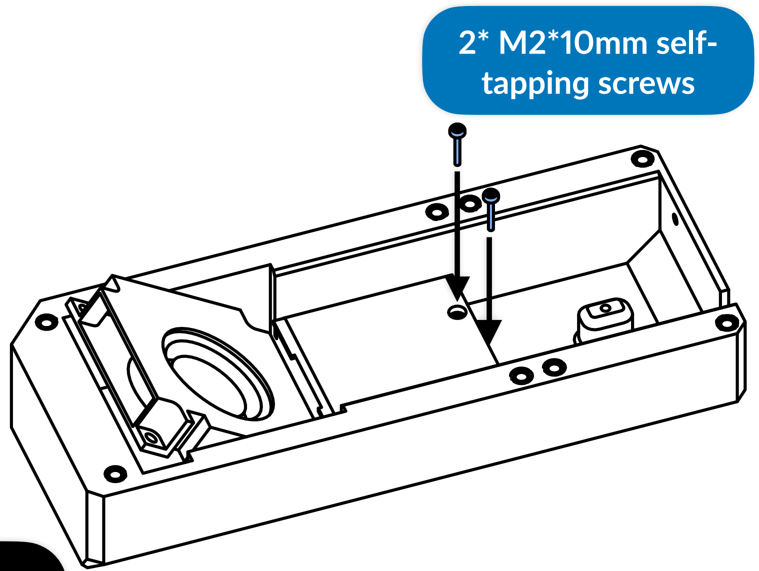
Bottom of the main unit

Note: this uses two-step procedure. Place the fan against the back. If the cable is too short, which usually is, it may need to be extended.
If the hybrid filament sensor is being installed, do not press the fan down just yet.



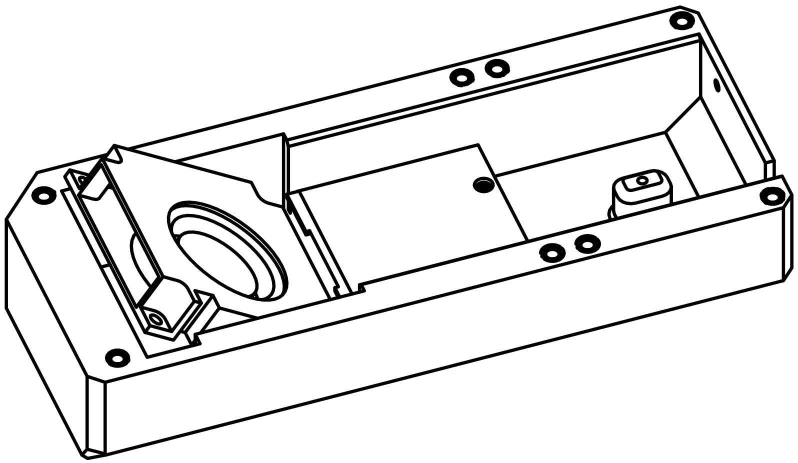
The fan will be installed like this, held by friction at the exhaust side and the clips at the back.





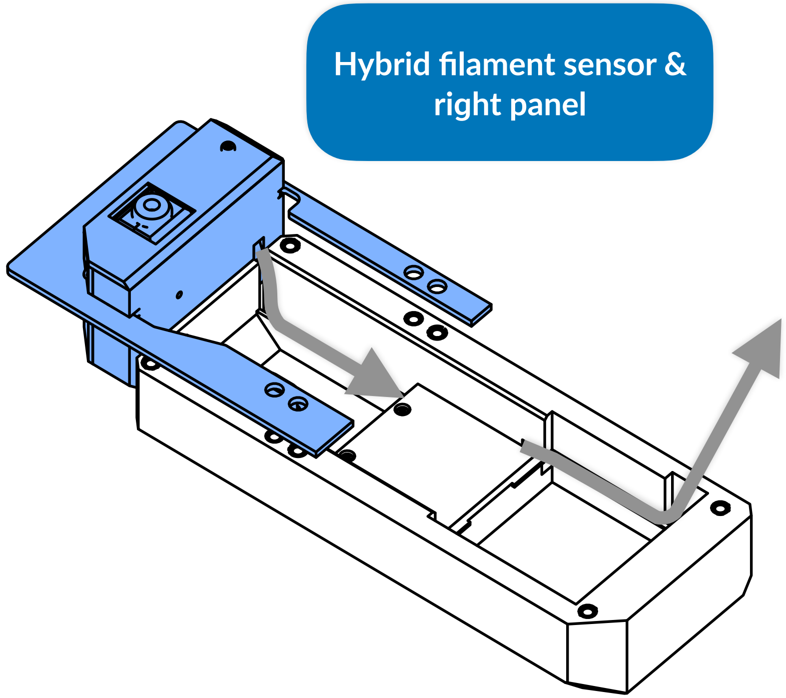
Bottom of the main unit

The reserved plate needs to be installed just to change the airflow. For the why it is called "reserved plate," stay tuned by following Oitswilliam in applicable platforms.

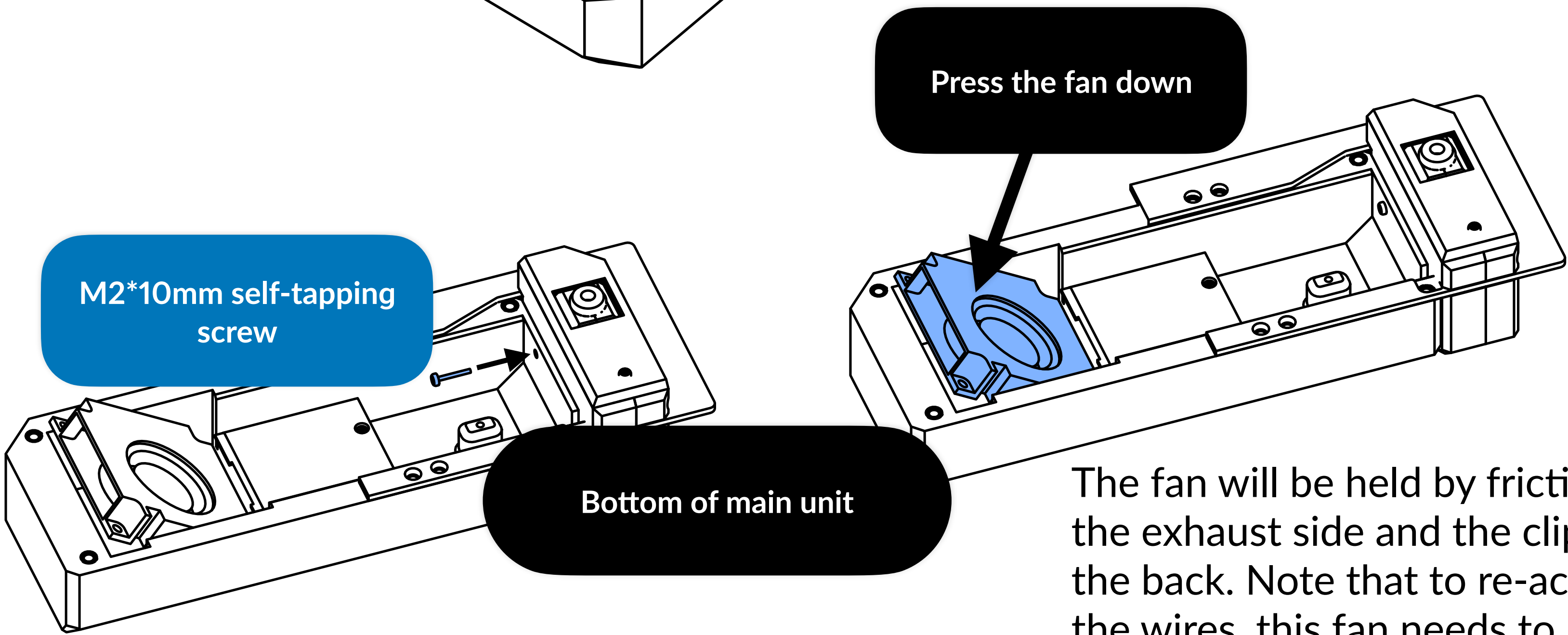


The bottommost part of the main unit is (mostly) complete.

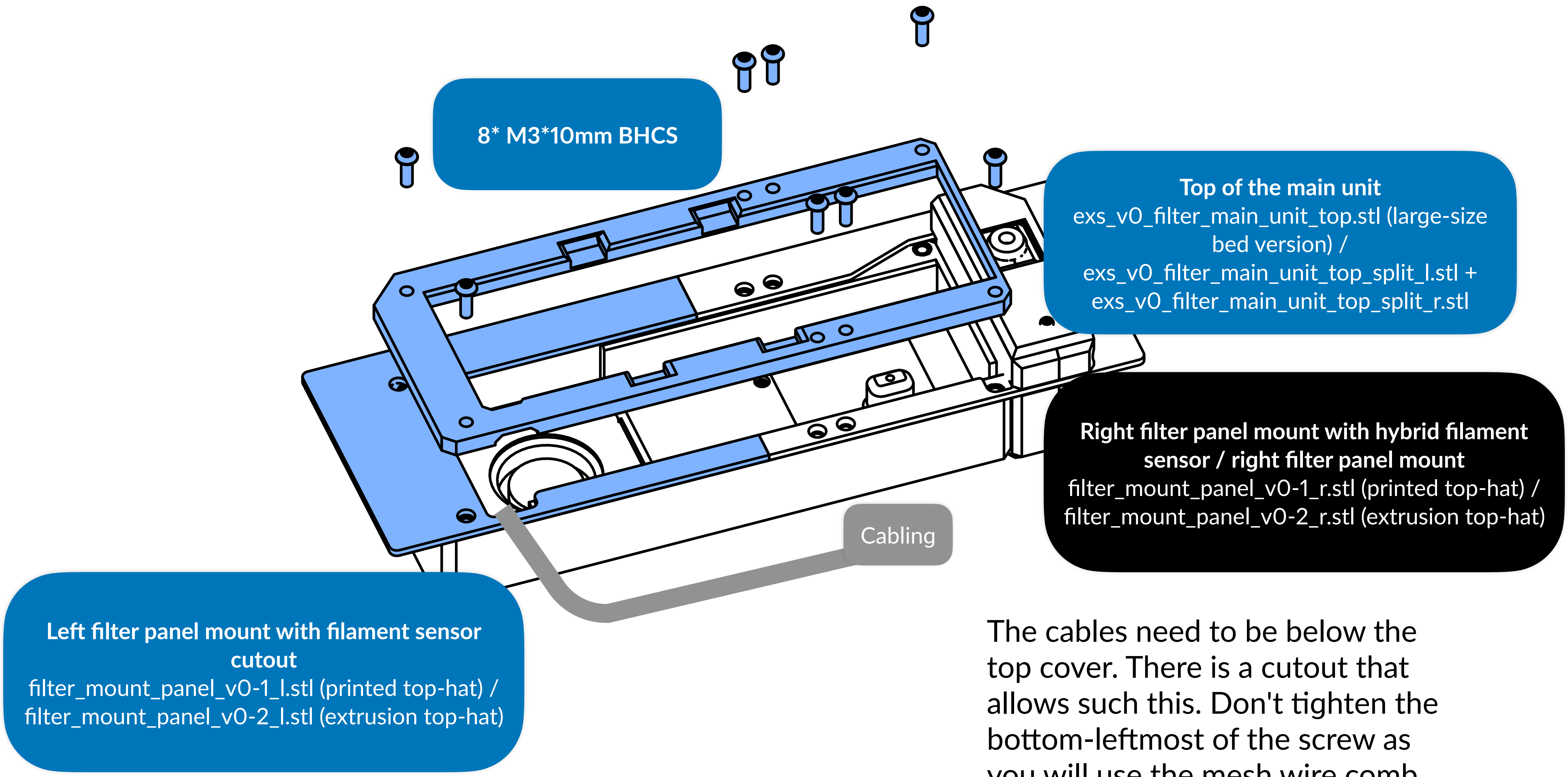
Crafting



Feed the cable from the hybrid filament sensor to the left cutout. **Note: the fan is hidden.** You may pull the fan out during installation.

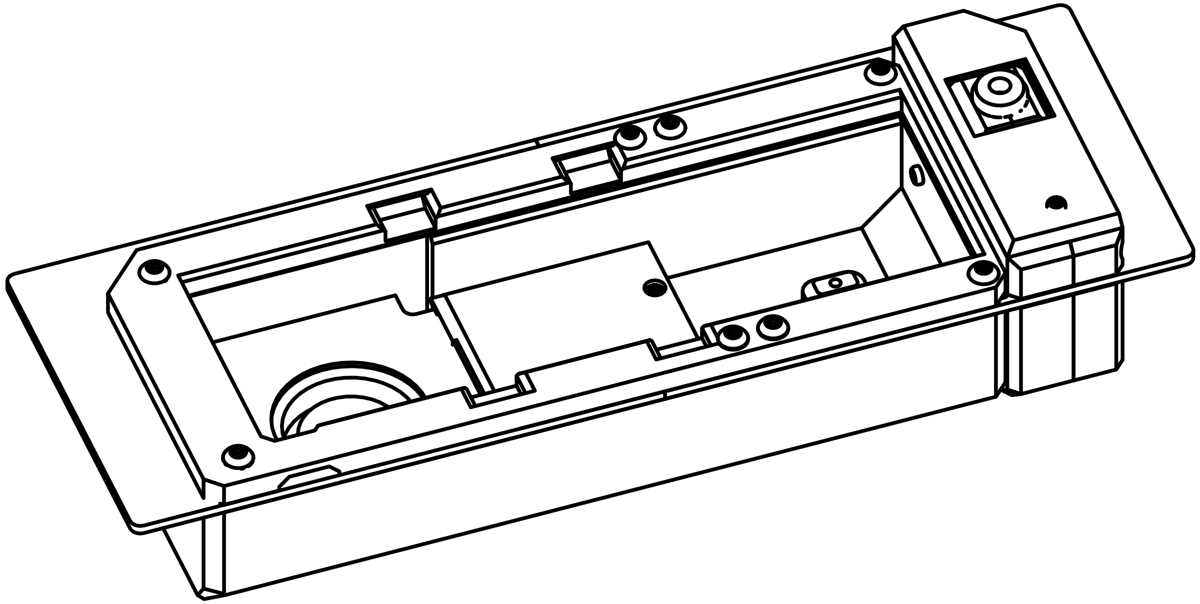


The fan will be held by friction at the exhaust side and the clips at the back. Note that to re-access the wires, this fan needs to be pulled first.



The cables need to be below the top cover. There is a cutout that allows such this. Don't tighten the bottom-leftmost of the screw as you will use the mesh wire comb.

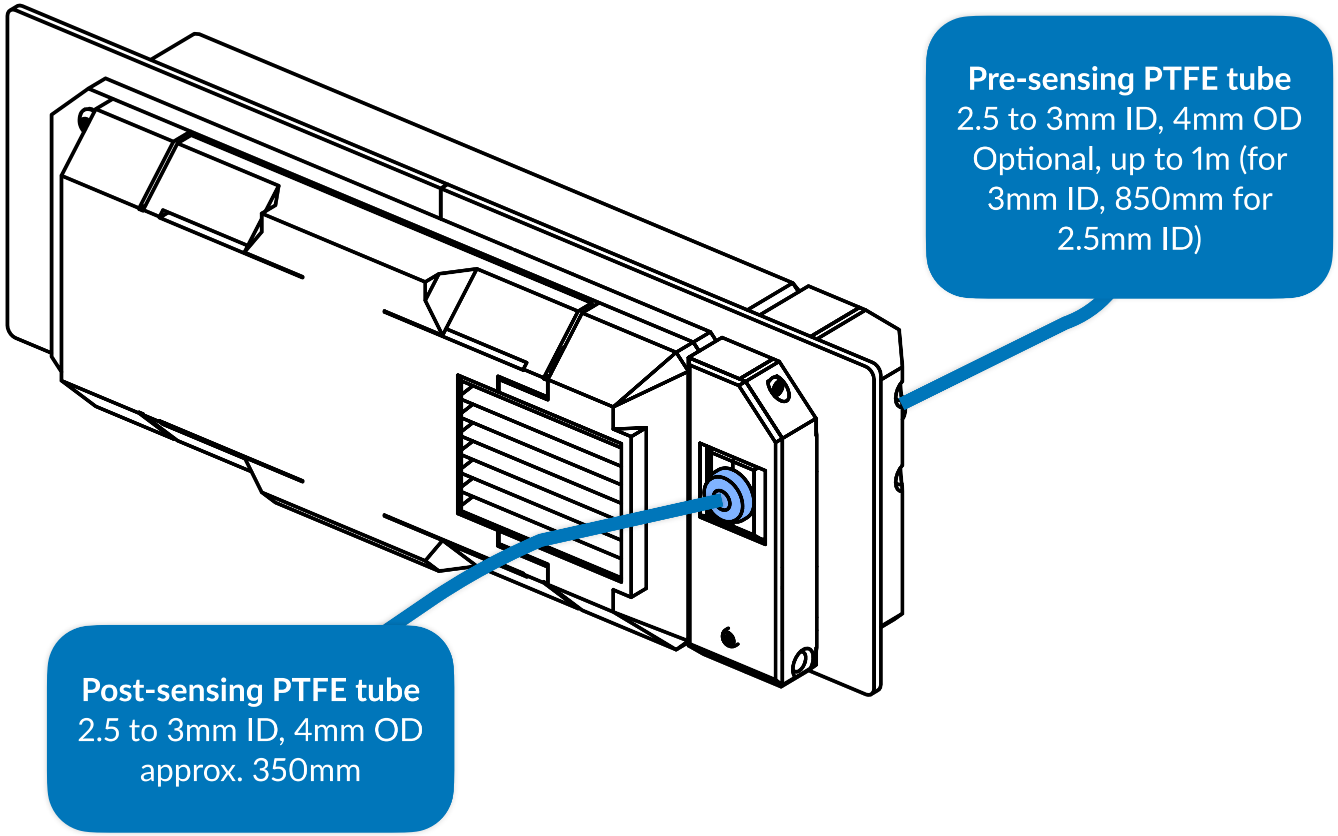
Once crafted, the wires need to be covered by the mesh comb. After that, the bottom-leftmost screw can be tightened.



The unit can then be mounted to the back of the printer. This replaces the back top-hat panel of Voron 0. Please refer to page 244 of the original Voron 0.2r1 manual or page 152 of the original Voron 0.1 manual. Note that top-hat from Voron 0.0 is not compatible.

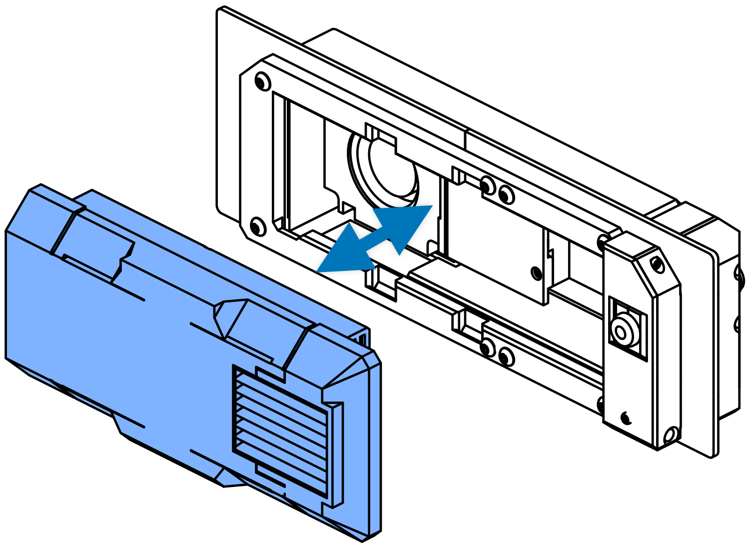
Reverse Bowden tube / PTFE tube attaching

To attach, press the tube against the hole of the coupler.
To remove the tube, press the collet of the coupler, and then pull the tube.
There are two couplers in the hybrid filament sensor, pre-sensing and post-sensing sides. Attach the post-sensing PTFE tube where the filament goes to the tool head, and optionally the pre-sensing PTFE tube where it attaches to the compatible dry box.

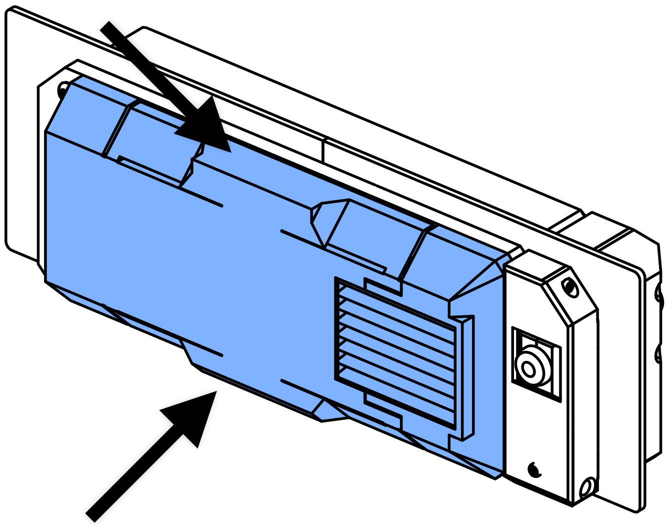


Attaching and removal

- To attach the cartridge to the main unit, press the cartridge to the main unit and hear a click.
- To remove the cartridge, squeeze the tabs of the cartridge and then pull it. Don't apply excessive force.



Attaching and removal directions



Tab squeezing for removal

Activated carbon pellets & front and back HEPA filters replacement

The filters for EXS are easy to be replaced, with minimal to without any tools.

- Remove the cartridge back and back HEPA filter security panels to allow replacement of the back HEPA filter (page 7 for reference),

- Remove the cartridge front bracket by pulling it to allow replacement of the front HEPA filter (page 8 for reference).
- Generally, the carbon pellets are recommended to be replaced on every 50 print hours, the front HEPA filter can be replaced on every 360 print hours and the back HEPA filter can be replaced on every 1,080 print hours. Note that the fan overheating is a late indicator of front HEPA filter having captured excessive particles.

Software configuration

The configuration for EXS is very simple unless if the hybrid filament sensor is added.

- The fan in EXS can be connected to the normal fan header of the microcontroller unit (MCU), but the hybrid filament sensor requires two digital signals to be assigned.
- Configuration for BTT Manta M4P demonstrates the connection of FAN2A / 2B (sometimes 3B) as chamber fan and using BL-Touch pins for filament presence and tangle sensing.

Troubleshooting

- The tangle sensor keeps triggering even though there are no tangles
  - **Culprit 1** - the tube before sensing is too long.
    - **Solution 1** - the tube before pre-sensing must be at least 2.5mm inner diameter and is recommended to be up to 850mm in this inner diameter, or 1 meter for tubes in 3mm inner diameter. Tubes in 2mm or under before the presence sensor is not recommended. Solution 2 may be used if you want to keep the length.
    - **Solution 2** - if you are using with the multiplexer, or want to keep the original length of the tube, a feed assistant is required and an additional buffer to work around is recommended. It should preload the material when the buffer runs out.
  - **Culprit 2** - tangle sensor's spring tension is too weak. Change the spring with the coil thickness that meets the spec, such as material of stainless steel and dimensions of 10*25*0.7mm, where 0.7mm is the coil thickness. If the spec spring's coil thickness unfortunately still false-triggers, you can then try using in coil thicknesses 0.1 to 0.3mm thicker. The maximum coil thickness that is allowed is 1mm, which result in total dimensions of 10*25*1mm. Using springs with coil thicknesses too thick may result in stalling the extruder motor instead (when tangle is caught), and 1mm coil thickness can already stall the motors from some of the extruder setups.
- The tangle sensor is not triggering
  - **Culprit 1** - the material is too soft, it might be TPU / PEBA with shore hardness lower than 95A.
    - **Solution 1** - bypass the hybrid sensor, this is because the materials that are this soft can lead most extruders stop extruding properly even with low frictions or tubes in high inner diameter.
    - **Solution 2** - add the feed assistant and buffer, this allows to preload the material and provide friction points opposite to usual.
  - **Culprit 2** - the lever for the tangle switch is missing. The lever for that switch is required.

- **Culprit 3** - bad wiring, check the wiring guide, be sure to have heat-shrink tubes with at least three types of color or labels.
- The filter is not filtering properly
  - **Culprit 1** - the activated carbon pellets has saturated. Replace the carbon pellets by first pulling the cartridge filter panel and then tilt down.
  - **Culprit 2** - the cartridge is using the non-compliant HEPA filters and / or have no foam around. The recommended filter grade is HEPA13 or over. If there is no 1mm foam around the sides, they should be wrapped around the sides.
- The fan is exhausting way too hot and too ineffectively.
- **Culprit** - the front HEPA filter has captured excessive particles. Replace the front HEPA filter. This is the late indicator of the front filter having captured excessive particles. Note that brushing the filter or capturing the particles back via vacuum cleaner is ineffective.

Assembly complete

And we are done! If you are satisfied with such product consider tagging @officialbunny350 on Facebook, @william86745 on Instagram or @realBunny350 on X (formerly known as Twitter) and share such product with us on Oitswilliam Pang Discord Server.

If you have enquires, problems or issues related to building or using such product, please report it on GitHub.com/Bunny350/OITSWILLIAMV0 and we will get on it.

Change log

- 2026-06-15 - General availability release. Add the required foam tape around the out side of inner HEPA filter.
- 2026-04-24 (alpha-ea) - Adjustments, this duplicates incompatibility note on before getting started, add troubleshooting section.
- 2026-04-08 (alpha-ea) - Adjustments, this changes the filter cartridge vectors to have snap-on clips instead of being secured via screws and make the document paper-compliant.
- 2026-03-28-0500 (alpha-ea) - Adjustments, such as fixing words. Add a section about PTFE tubes around the tool head and the pre-sensing. Fix incorrect change log date for the first change.
- **2026-03-28** (alpha-ea) - initial release, make exclusive to the followers of Oitswilliam Pang until June 15.

Oitswilliam Pang has introduced a dedicated Facebook page (2018) before Bambu Lab even existed (2020). I wished Bambu exist on somewhere 2009.

Manual designed by Oitswilliam Pang.
One-person operated design.
www.oitswilliam.com